Elements, compounds or mixtures? (1)

In science, it is important to know the difference between elements, compounds and mixtures. Try to explain what you think each of these words means:

1. An element is

_____

_____

2. A compound is

_____

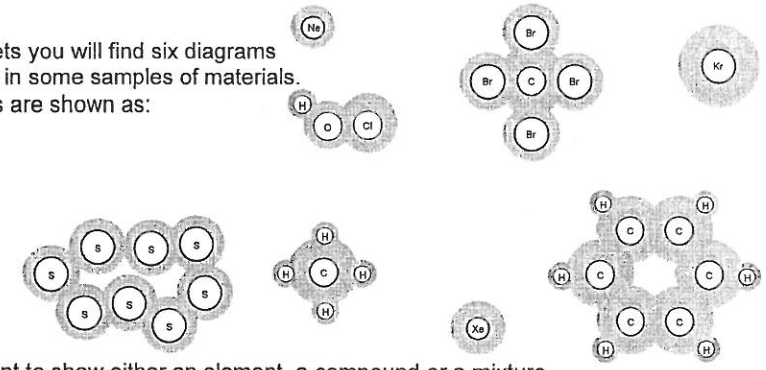
_____

3. A mixture is

_____

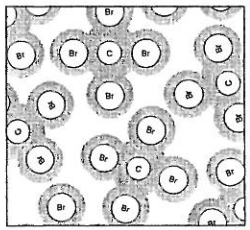
_____

On the following sheets you will find six diagrams showing the particles in some samples of materials. The different particles are shown as:



Each diagram is meant to show either an element, a compound or a mixture.

Decide whether each diagram represents an element, a compound, or a mixture, and explain your reasons.



4. This diagram shows particles in

_____

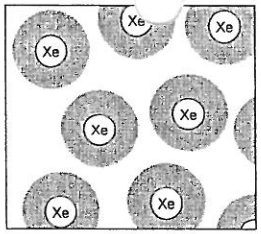
I think this because

_____

_____

_____

_____



5. This diagram shows particles in

_____

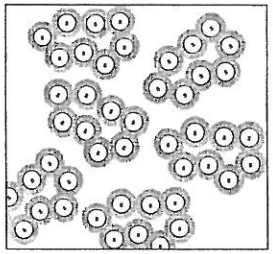
I think this because

_____

_____

_____

_____



6. This diagram shows particles in

_____

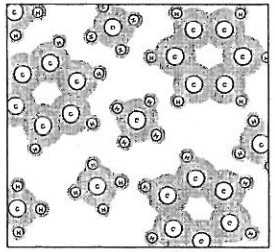
I think this because

_____

_____

_____

_____



7. This diagram shows particles in

_____

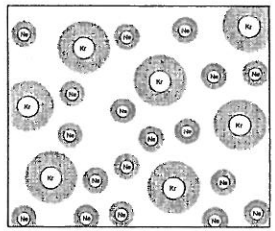
I think this because

_____

_____

_____

_____



8. This diagram shows particles in

_____

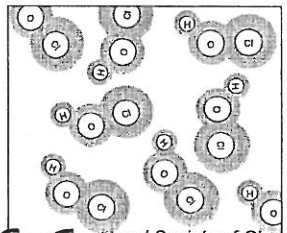
I think this because

_____

_____

_____

_____



9. This diagram shows particles in

_____

I think this because

_____

_____

_____

RS•C *Royal Society of Chemistry Student Sheets – Chemical Misconceptions – Elements probe*

Elements, compounds and mixtures (2)

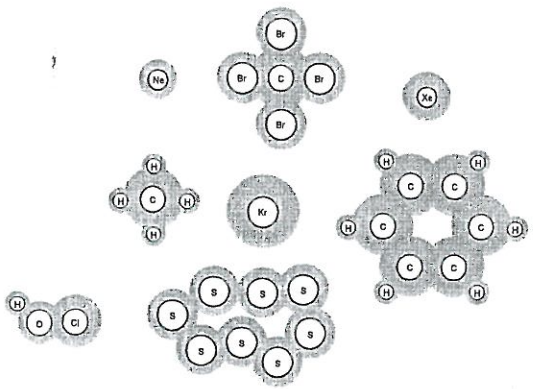
Pure substances and mixtures

In science, it is important to know the difference between pure substances and mixtures of several substances. Scientists think about the differences in terms of the particles which are in the materials.

Scientists believe that all matter (all solids, liquids and gases) is made up from tiny particles that are much too small to be seen.

The tiniest particles are given names like 'electron' and 'proton'. These are arranged into slightly larger (but still very tiny) particles called atoms, ions and molecules. In many materials the particles are called molecules.

Here are some pictures that scientists use to represent atoms and molecules.

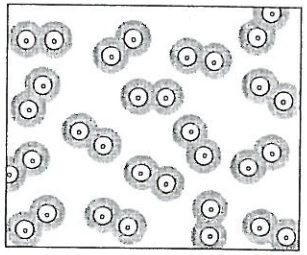


The letters are labels used by scientists to help identify the particles. There are many different types of atoms and molecules, and these pictures just show a few examples.

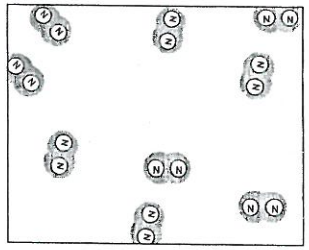
Different substances contain different molecules

The three diagrams below show two different substances. Which two diagrams show the same substance?

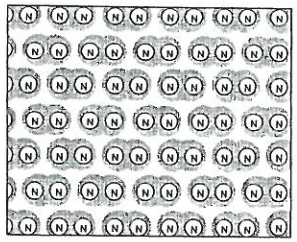
A



B



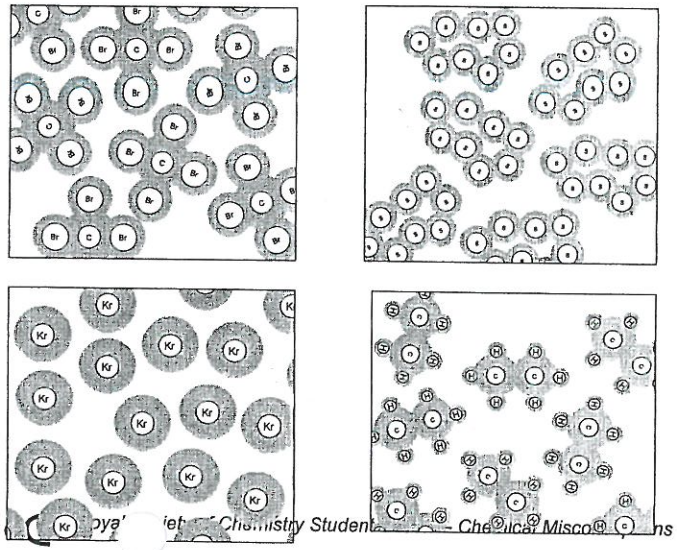
C

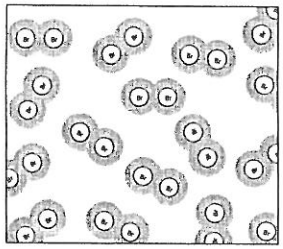


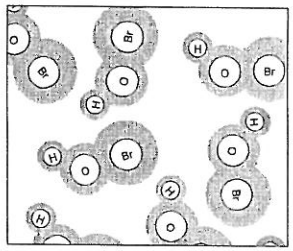
Answer

___ and ____ are the same substance because they contain the same molecules. They look different because they show how the molecules are arranged when the substance is a gas (B) and a solid (C). Diagram A shows a different substance, because it shows different molecules.

Some materials only contain one type of molecule or atom. These are called single substances (or pure substances).
The following diagrams show some single substances. They are each different substances because they have different types of molecules.





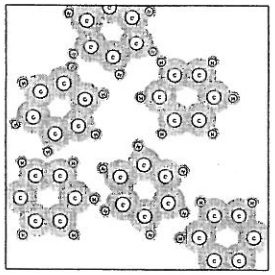


When substances are mixed, their molecules become mixed up. We call the new material a mixture.

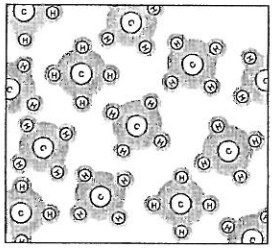
A mixture contains more than one type of atom or molecule.

The following diagrams show the molecules in two pure substances before mixing, and the mixture of molecules afterwards.

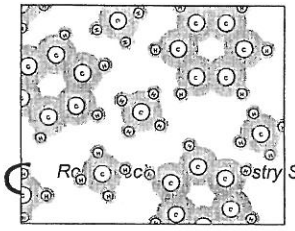
Look at the diagrams closely, and label each of them as either a single substance, or a mixture.



1. _____

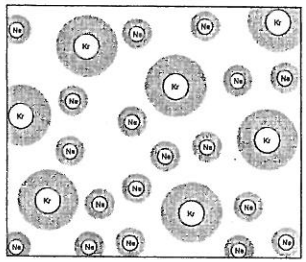


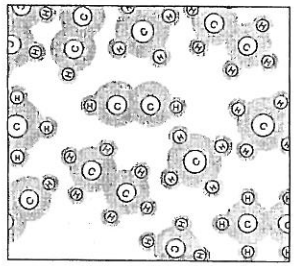
2. _____



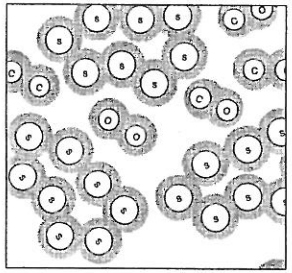
3. _____

Here are some other diagrams showing mixtures:





The following two diagrams show a single substance and a mixture. Complete the labels to show you know which is which:

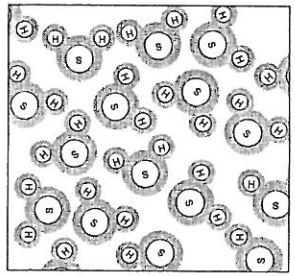


4. This diagram shows a

_____

because there is more than one type of

_____



5 This diagram shows a

_____

because there is only one type of

_____

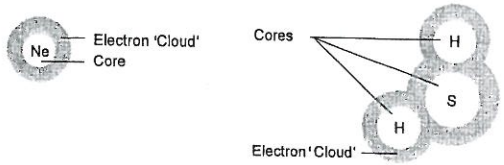
Two types of substance - element and compounds

To tell the difference between single substances and mixtures you need to be able to recognise diagrams of different types of atoms and molecules.

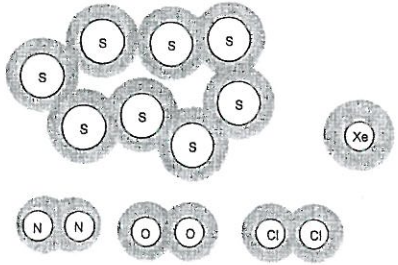
But scientists divide single substances into two types: elements and compounds. To spot the difference between elements and compounds you have to look more closely at the atoms or molecules.

Atoms are made of one core surrounded by a 'cloud' of electrons. Molecules are made of

two or more cores surrounded by a 'cloud' of electrons.

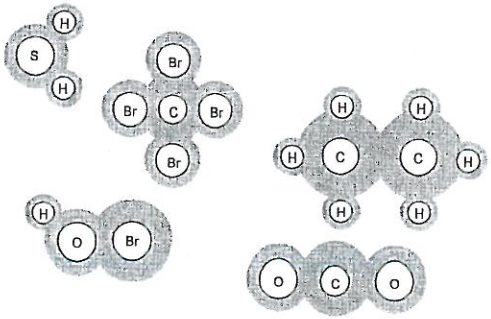


The atoms or molecules that form an element have only one type of core. (Scientists use different letters to represent the different elements).

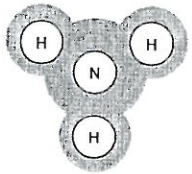


Atoms or molecules that have one type of core

In a compound the molecules have two or more different types of core.



The following two diagrams show a molecule of an element and a molecule of a compound. Complete the labels to show you know which is which:

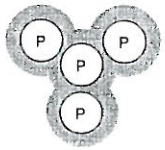


6. This diagram shows

_____

because there is more than one type of

_____



7 This diagram shows

_____

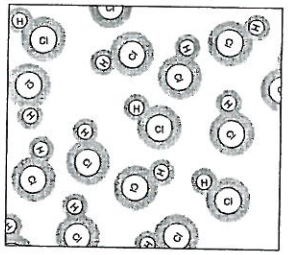
because there is only one type of

_____

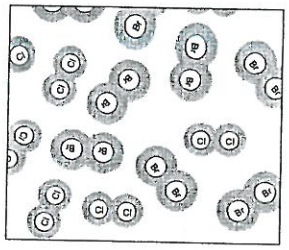
To summarise:

Looking at:	If the same:	If different:
The types of single uncombined atoms	A single substance	A mixture
The types of molecules	A single substance	A mixture
The types of cores in a single molecule	A molecule of an element	A molecule of a compound

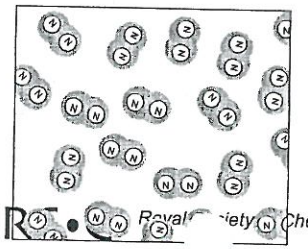
The following diagrams show molecules in a mixture, element and compound. Complete the labels to show you know which is which:



8. _____



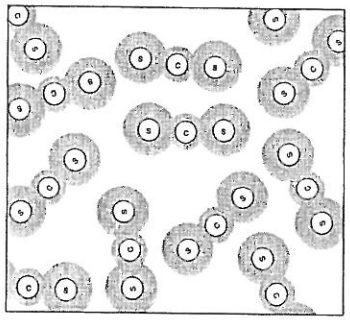
9. _____



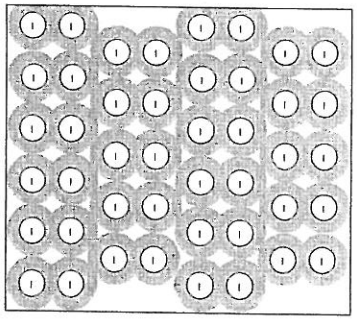
10. _____

Elements, compounds or mixtures? (3)

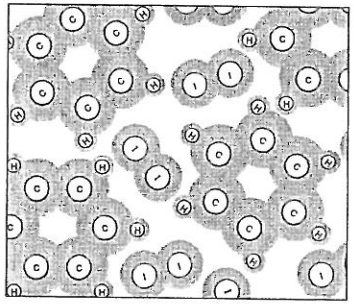
Each of the following diagrams show the particles in a material. For each diagram, write whether you think it represents an element, a compound or a mixture



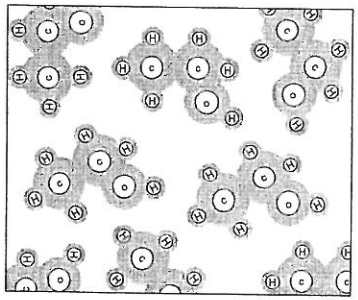
1. _____



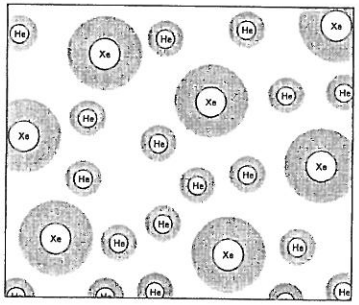
2. _____



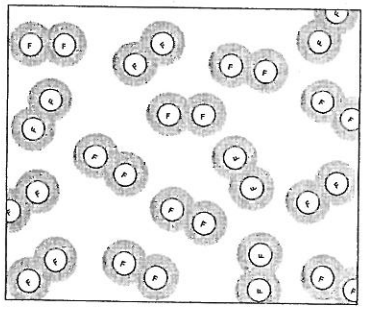
3. _____



4. _____



5. _____



6. _____

Element, Compound or Mixture

Rust is formed when iron objects are left out in the air. The iron reacts with the oxygen in the air, forming iron oxide. Rust can be turned back into iron, by mixing it with carbon and heating it. I think rust is _____ (an element / a mixture / a compound) because _____

_____

There are a number of different gases in the air. The main ones are nitrogen and oxygen, but there are also small amounts of argon, carbon dioxide and water vapour. I think air is _____ (an element / a mixture / a compound) because _____

_____

_____

Helium is a gas. There are tiny amounts of helium in the air. Helium does not burn, and will not react with any other chemicals. It cannot be split up by chemical reactions. It think helium is _____ (an element / a mixture / a compound) because _____

_____

_____

Evaporating salty water results in common salt being left behind. This salt is called sodium chloride, and is made up of two elements, sodium (Na) and chlorine (Cl). I think salt is _____ (an element / a mixture / a compound) because _____

_____

_____

Now go back over your work, underlining any element in red; any compound in green; and any mixture in blue. Stick the worksheet in when you are done...